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Flow Field and Performance Characteristics of Combustor Diffusers: A Basic Study

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Results of a detailed study concerning the influence of geometric as well as fluid mechanic parameters on the performance of a plane model combustor diffuser in cold flow are presented. For a qualitative insight into the complex flow field inside the prediffuser, the sudden expansion region, and the flow field around the flame tube dome, results of a flow visualization study with the hydrogen bubble method as well as with the ink jet method are presented for different opening angles of the prediffuser and for different flame tube distances. Also, quantitative data from detailed measurements with LDV and conventional pressure probes in a geometrically similar air-driven setup are presented. These data clearly demonstrate the effect of boundary layer thickness as well as the influence of different turbulence levels at the entry of the prediffuser on the performance characteristics of combustor diffusers. The possibility of getting an unseparated flow field inside the prediffuser even at large opening angles by appropriately matching the diffuser's opening angle and the flame tube distance is demonstrated. Also, for flows with an increased turbulence level at the entrance—all other conditions held constant—an increased opening angle can be realized without experiencing flow separation. The comparison of the experimental data with predictions utilizing a finite-volume-code based on a body-fitted coordinate system for diffusers with an included total opening angle less than 18 deg demonstrates the capability of describing the flow field in combustor diffusers with reasonable accuracy.

Introduction

One of the most challenging tasks in the design of future generations of gas turbine engines is to minimize the pollutant emissions and to maintain at least identical lifetime as well as the reliability of the entire engine at improved efficiency. In order to be successful, the combustion process inside the combustor's flame tube has to be controlled effectively even under the extreme conditions of high temperature and high pressure. With its dominating effect on the flow field inside and outside the flame tube, the combustor diffuser shows considerable potential in optimizing gas turbine engines. For reasons of high efficiency, the combustor diffuser has to decelerate the flow with minimum pressure loss, resulting in a pressure level in the region surrounding the flame tube as high as possible to ensure deep penetration of the air jets into the flame tube. Also, for reasons of reliability, the combustor diffuser has to provide similar flow profiles at the diffuser's exit under a wide range of operating conditions. The factor limiting the performance of diffusers is the onset of flow separation, which is influenced—among other parameters—by the length and the opening angle of the diffuser. The diffuser length is an even more important parameter in the design of jet engines than in the design of stationary gas turbine engines because of its effect on engine weight and rotor stiffness. Modern gas turbine combustor diffusers are of relatively short design comprising a prediffuser to reduce the air velocity followed by a sudden expansion. Whereas for conventional diffusers performance charts exist (see, e.g., Reneau et al., 1967), little information is available for the correlation of the geometric and fluid dynamic parameters and the corresponding performance parameters for combustor diffusers. In comparison with conventional diffusers, additional parameters exist influencing combustor diffuser's performance, especially the shape and the position of the flame tube relative to the prediffuser, the width of the flame tube annuli surrounding the flame tube, as well as the position and size of mixing holes along the flame tube's wall.

Up to now, relatively few studies can be found that consider these parameters systematically. In general, the design is based on the well-known results of Reneau et al. (1967), Kline et al. (1969), Adkins (1983), and Adkins and Wardle (1992). Earlier studies of the characteristics of annular diffusers were presented by Ainley (1945), Johnston (1953), and Howard et al. (1967), but without taking into account flame tube effects. Biaglow (1971) could demonstrate the advantage of dump-type diffuser configurations compared to faired diffusers. He showed that the temperature distribution at the exit of a two-dimensional model of a dump diffuser is insensitive to the inlet velocity profile. In an extensive study, Fishenden and Stevens (1974) examined the influence of the prediffuser geometry, the flow split between the two annuli surrounding the flame tube, and the distance between the flame tube and the prediffuser exit. The largest area ratio of the prediffusers under examination was 1.8. Klein (1988) in his work showed that the radial velocity profile more than the circumferential velocity profile determines the performance characteristics of combustor diffusers. Wagner et al. (1980) investigated the effects of flow splits on the performance of a curved wall dump diffuser with a canted prediffuser. More recently, Srinivasan et al. (1990) published the results of an experimental study concerning the effects of flame tube distance, flow split, prediffuser inlet velocity profile and combustor channel height on diffuser performance in a 60 deg sector element of a combustor diffuser. Despite the increasing efforts to obtain more experimental data to describe the correlation between geometric, fluid dynamic, and diffuser performance parameters, there is still a lack of accurate data. A comparison of the available results is difficult because of the different geom-

Contributed by the International Gas Turbine Institute and presented at the 39th International Gas Turbine and Aeroengine Congress and Exposition, The Hague, The Netherlands, June 13–16, 1994. Manuscript received by the International Gas Turbine Institute February 19, 1994. Paper No. 94-GT-212. Associate Technical Editor: E. M. Greitzer.

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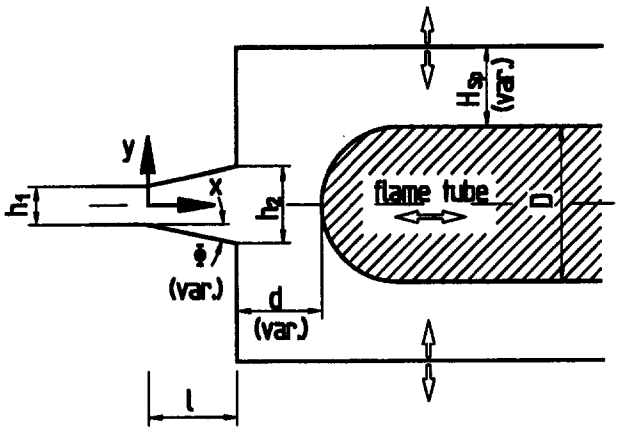

Fig. 1 Sketch of the model diffuser

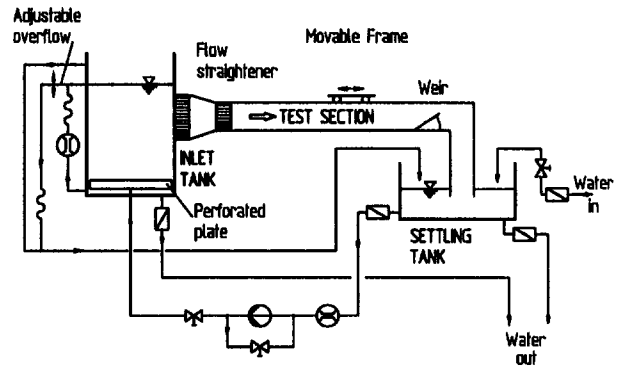

Fig. 2 Water test facility

etries and fluid dynamic boundary conditions. Sovran and Klomp (1967) basically show the correlation between the performance parameters of plane, conical, and annular diffusers. However, the effect of flame tubes is not considered. Therefore, up to now it is not possible to generate a general performance chart for combustor diffusers, comparable to those for plane diffusers (e.g., Reneau et al., 1967).

A solution in the design of diffusers for gas turbine combustor applications is found in the use of numerical codes to predict the flow field. The possibility to calculate the flow field inside complex geometries has been shown by us as well as by various authors (see, e.g., Shyy, 1985; Noll, 1986; Koutmos and McGuirk, 1989; Noll et al., 1989; Noll and Wittig, 1991). To verify and to improve available computer codes, detailed experimental data with well-defined boundary conditions are required. In extending our earlier studies of diffusers (Kim, 1976), two test facilities were developed. Both are equipped with plane model combustor diffusers. The plane arrangement has the advantage of good accessibility and flexibility and in addition is sufficient to demonstrate the flow behavior in annular dump diffusers, as was shown by Little and Manners (1993) in a recently published paper. In the air-driven test rig, detailed measurements with LDV as well as with conventional instrumentation were performed. A description of the setup and the measuring techniques used is presented elsewhere (Hestermann et al., 1991). The second test facility is used mainly for flow visualization studies and uses water as the fluid. In the following, a description of the water test facility and the techniques used for flow visualization is presented along with the results. Also some results of the LDV measurements in the air-driven setup are presented.

Experimental Setup

The diffuser model under investigation in the water test facility is geometrically similar to that used in the air-driven test rig. It comprises a straight-walled prediffuser followed by a dump diffuser and a flame tube, as shown in Fig. 1. All components were made of plexiglass to enable a great variety of flow visualization techniques. Similar to the air-blown test facility, a great number of parameters can be varied independently. In this paper the visualized flow fields for different opening angles of the prediffuser and different flame tube distances are presented. To allow direct comparisons to real engine flow conditions, the experiments were conducted at Reynolds numbers, based on inlet height h_1, of about $\mathrm{Re}_{h_1} \sim 0.4 \times 10^5$. This is a good approximation of real engine conditions.

The water test facility is a closed circuit arrangement as shown in Fig. 2. The main components are the inlet tank, the test section, and the settling tank. It can be operated either with a free surface or as a covered tunnel. The cross section of the test section is 600 × 280 mm with a length of 1800 mm. The walls consist of glass to enable the use of light sheet techniques as well as of other flow visualization methods. The excellent optical access enables observation of the complete flow field all along the length of the test section. To avoid mechanical vibrations, the test section is mounted between two compensators. The flow rate can be controlled by a throttle valve and a bypass. At the exit of the test section, a weir is mounted to adjust the water height inside the test section. To measure the flow rate, a vortex meter is installed in the pipe system. The flow rate through the adjustable overflow can be measured by an additional flow meter, if necessary.

Inside the test section, a contraction module is mounted, followed by a parallel-walled entry section. To ensure turbulent flow conditions, trip wires with a thickness of 5 percent of the channel height are installed on both side walls of the entry section. The diverging diffuser walls are attached to the entry section via two cylinders allowing the adjustment of any opening angle on either side.

Flow Visualization Techniques

For the visualization of the complex flow field, two techniques have been applied: To give an impression of the entire flow field, the hydrogen bubble method is employed. In those regions where it was desirable to get additional information

Nomenclature

AR = area ratio = A_2/A_1
B = blockage factor = $1 - 1/A \times \int \frac{u}{u_{\max}} dA$
$C_{p,R}$ = pressure recovery coefficient
D = flame tube height
d = flame tube distance
d_b = diameter of the turbulence generating bars
h = channel height
H_{Sp} = annulus height
l = prediffuser length
Tu = turbulence intensity
x, y = coordinates
Φ = half-angle of the prediffuser

Subscripts

1 = reference position; $x/h_1 = -2.0$
2 = exit of prediffuser
VD = measured from reference position to prediffuser exit
ges = measured from reference position to the exit of the flame tube annuli

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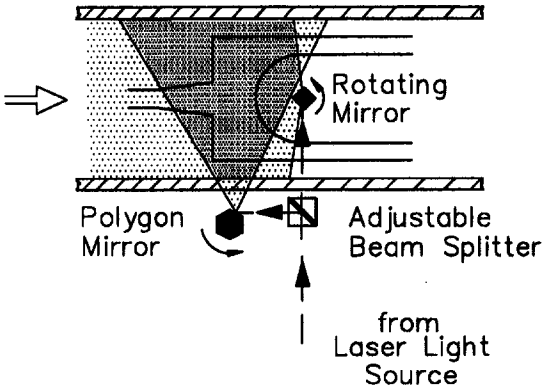

Fig. 3 Laser light sheet setup (top view)

on the flow, the ink jet method was utilized. Two different illumination techniques were used. For the hydrogen bubble method, a laser light sheet was generated by use of two rotating mirrors. The mirror positioned inside the flame tube (see Fig. 3) was encapsulated in a waterproof plexiglass box to illuminate the flow field inside the prediffuser. With the ink jet method, a standard photographic light source and a reflecting white plate were employed (see Fig. 4) resulting in a dark image of the ink in front of a light background.

Photographs of the flow field were taken by a standard camera as well as with a video camera. The application of the hydrogen bubble method made it necessary to use highly sensitive films even with a laser output power of about 2 W. Both cameras can be mounted on a movable frame to allow positioning of the cameras at any position at the top of the test section.

Results

As a result of earlier studies in an air-blown test facility (Hestermann et al., 1991) it was shown that the flow inside the prediffuser has a dominating effect on the flow field of the entire combustor diffuser. On the other hand, the geometry downstream of the prediffuser affects the flow inside the prediffuser. Applying the nonintrusive LDV measuring technique, it could be shown that for moderate diffuser opening angles of $2\Phi = 15$ deg, the velocity distribution at the prediffuser's exit, starting with a nozzlelike profile for large flame tube distances, flattens with decreasing flame tube distances. As a consequence, the flow near the prediffuser walls at its exit exhibits normal components toward the upper and lower walls, respectively, and the velocity near the walls close to the prediffuser exit increases, resulting in a stabilization of the flow with decreasing flame tube distance. For moderate opening angles the flow is attached throughout the prediffuser. The results of the flow

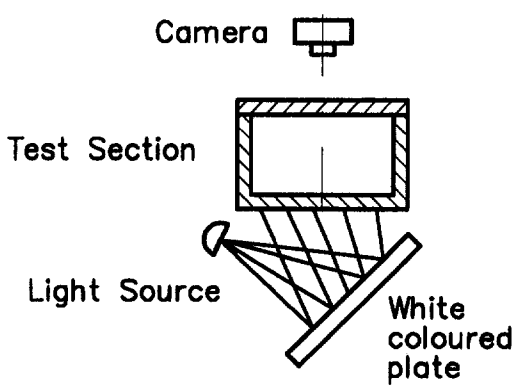

Fig. 4 Ink jet method: illumination

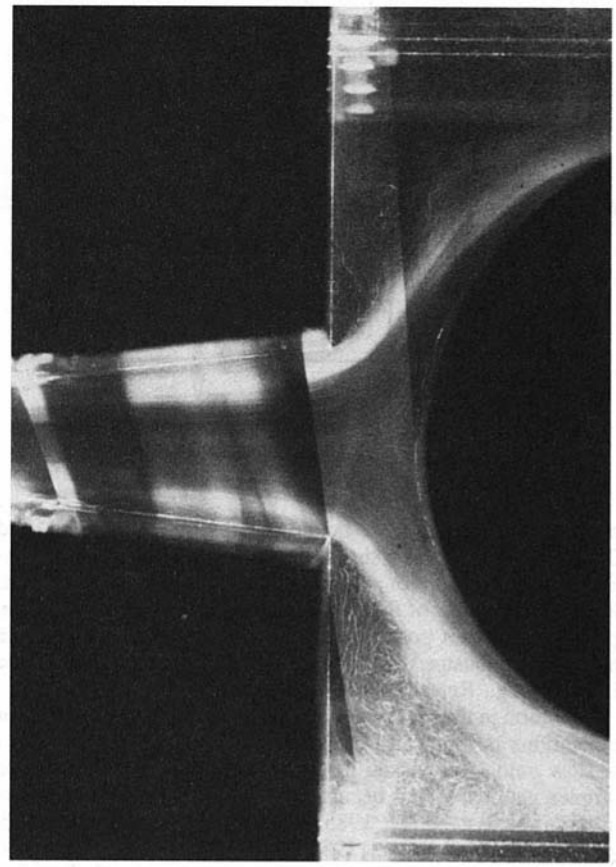

Fig. 5 Hydrogen bubble flow visualization (geometric data: $2\Phi = 15$ deg; $l/h_1 = 3.0$; $H_{Sp}/h_1 = 1.0$; $d/h_1 = 0.74$)

visualization confirm these quantitative results qualitatively, as shown in Figs. 5 to 7. Clearly visible is the attached flow inside the entire prediffuser, the flow separation at the edge of the prediffuser's exit and the reattachment of the flow downstream of the dump diffuser.

At small flame tube distances ($d/h_1 = 0.74$, see Fig. 5), the upstream influence of the flame tube dome inside the prediffuser is indicated by the curvature of the hydrogen bubbles' path lines starting with the curvature already inside the prediffuser. This is in good agreement with the results of the LDV measurements presented earlier. At large flame tube distances ($d/h_1 = 2.51$, see Fig. 7) the flow propagates similarly to a jet for a certain

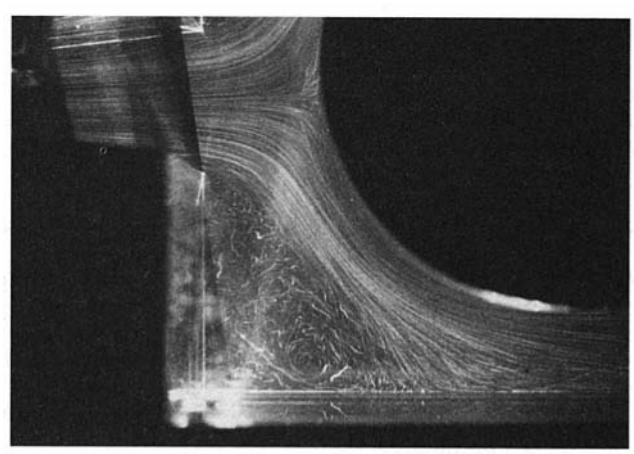

Fig. 6 Hydrogen bubble flow visualization (geometric data: $2\Phi = 15$ deg; $l/h_1 = 3.0$; $H_{Sp}/h_1 = 1.0$)

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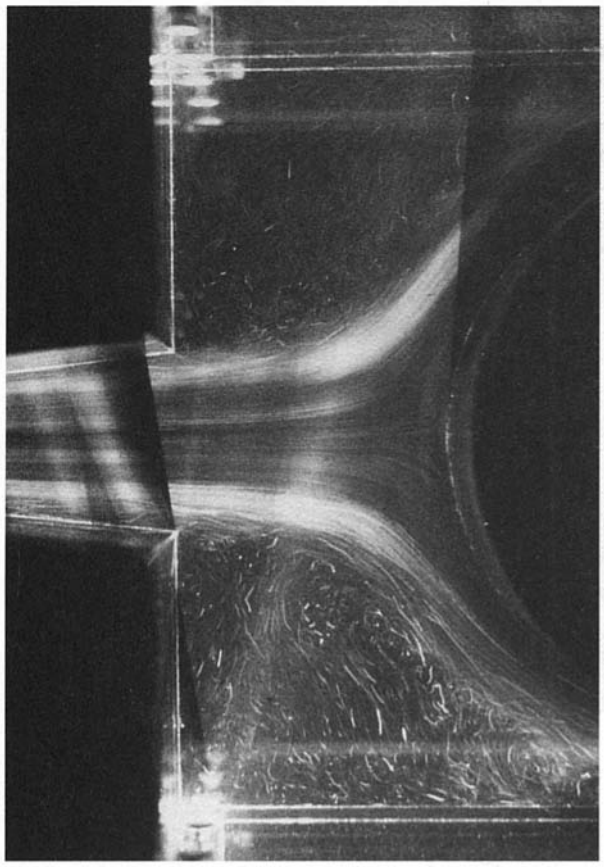

Fig. 7 Hydrogen bubble flow visualization (geometric data: $2\Phi = 15$ deg; $l/h_1 = 3.0$; $H_{Sp}/h_1 = 1.0$; $d/h_1 = 2.51$)

distance downstream of the prediffuser's exit before it is influenced by the flame tube dome. In the corner region of the dump diffuser, a clearly visible vortex exists as expected. A much smaller counterrotating vortex can be recognized at the edge of the prediffuser's exit. At a moderate flame tube distance of $d/h_1 = 1.26$ (see Fig. 6) the deflection of the flow starts directly at the prediffuser's exit without any upstream effect of the flame tube dome visible inside the prediffuser. Obviously, the flow deflection is not affected by the edge at the prediffuser's exit. This is a further demonstration for the existence of an optimum pressure recovery at the moderate flame tube distances mentioned by Fishenden and Stevens (1974). Also visible in the photographs is the location of the reattachment point at the casing walls of the dump diffuser, varying with flame tube distance.

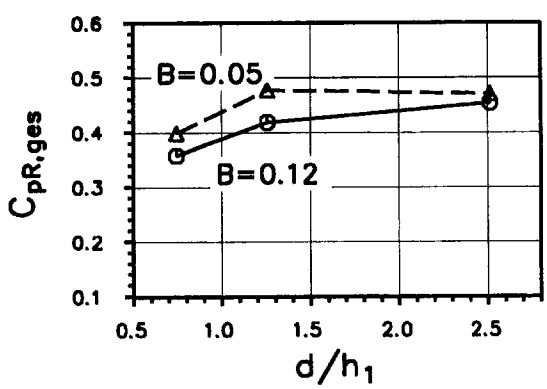

Fig. 8 Effect of boundary layer thickness on overall pressure recovery (B = inlet blockage factor); geometric data: $2\Phi = 15$ deg; $l/h_1 = 3.0$

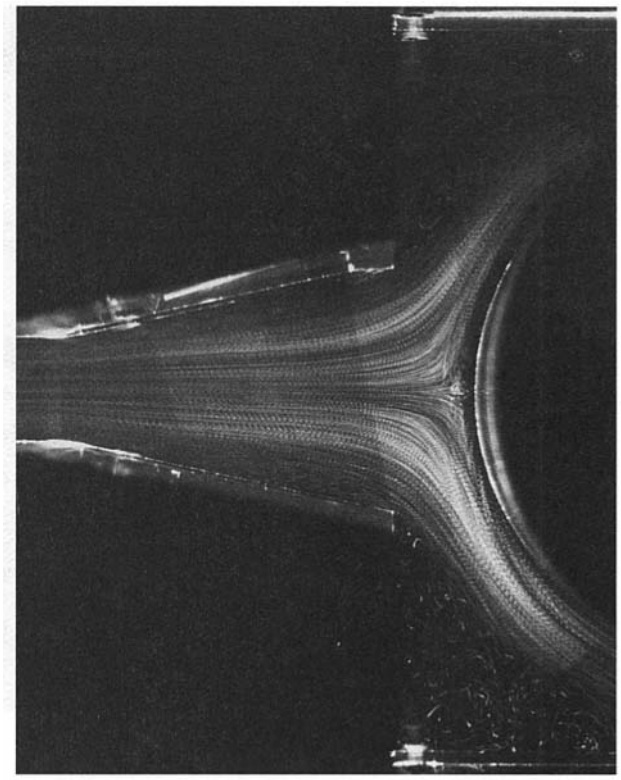

Fig. 9 Hydrogen bubble flow visualization (geometric data: $2\Phi = 25$ deg; $l/h_1 = 3.0$; $H_{Sp}/h_1 = 1.0$; $d/h_1 = 0.74$)

The photographs shown here have been taken for thin boundary layers at the entrance of the prediffuser. As the results of the LDV measurements inside the prediffuser show (see Hestermann et al., 1991), the strong upstream effect of the flame-tube on the velocity profile at the prediffuser's exit decreases with increasing boundary layer thickness at the prediffuser's entry because of the larger momentum in the core region at thicker boundary layers and at equal mass flow rates. Also, the overall pressure recovery, measured from the entry of the prediffuser to the exit of the flame tube annuli, decreases with increasing boundary layer thickness (see Fig. 8).

In real engines the boundary layer thickness at the diffuser's entry is given by the compressor's discharge flow. So from a designer's point of view, the only way to increase the pressure recovery at a given diffuser length is to increase the opening angle. In doing so, the flow remains attached to both prediffuser walls for moderate opening angles. At an opening angle of $2\Phi = 18$ deg and at large flame tube distances ($d/h_1 = 2.51$), the flow starts separating from the prediffuser's diverging walls for short time intervals. With increasing opening angle, the time intervals showing flow separation grow and the extension of the separated regions increases similarly to the flow phenomena described by Reneau et al. (1967) for conventional two-dimensional diffusers without flame tube. This unsteady separation can be eliminated by decreasing the flame tube distance. The larger the opening angle, the smaller the flame tube distance has to be to suppress flow separation. At a total opening angle of $2\Phi = 22$ deg, the flow separation inside the prediffuser can only be prevented for flame tube distances $d/h_1 < 1$. For larger total opening angles ($2\Phi = 25$ deg) flow separation cannot be prevented even at small flame tube distances. Especially at these large opening angles, the flow stabilizing effect of the flame tube can clearly be demonstrated: At large opening angles and simultaneously at large flame tube distances the flow always separates from one diverging wall inside the prediffuser while the flow on the opposite side remains attached. Thereby two

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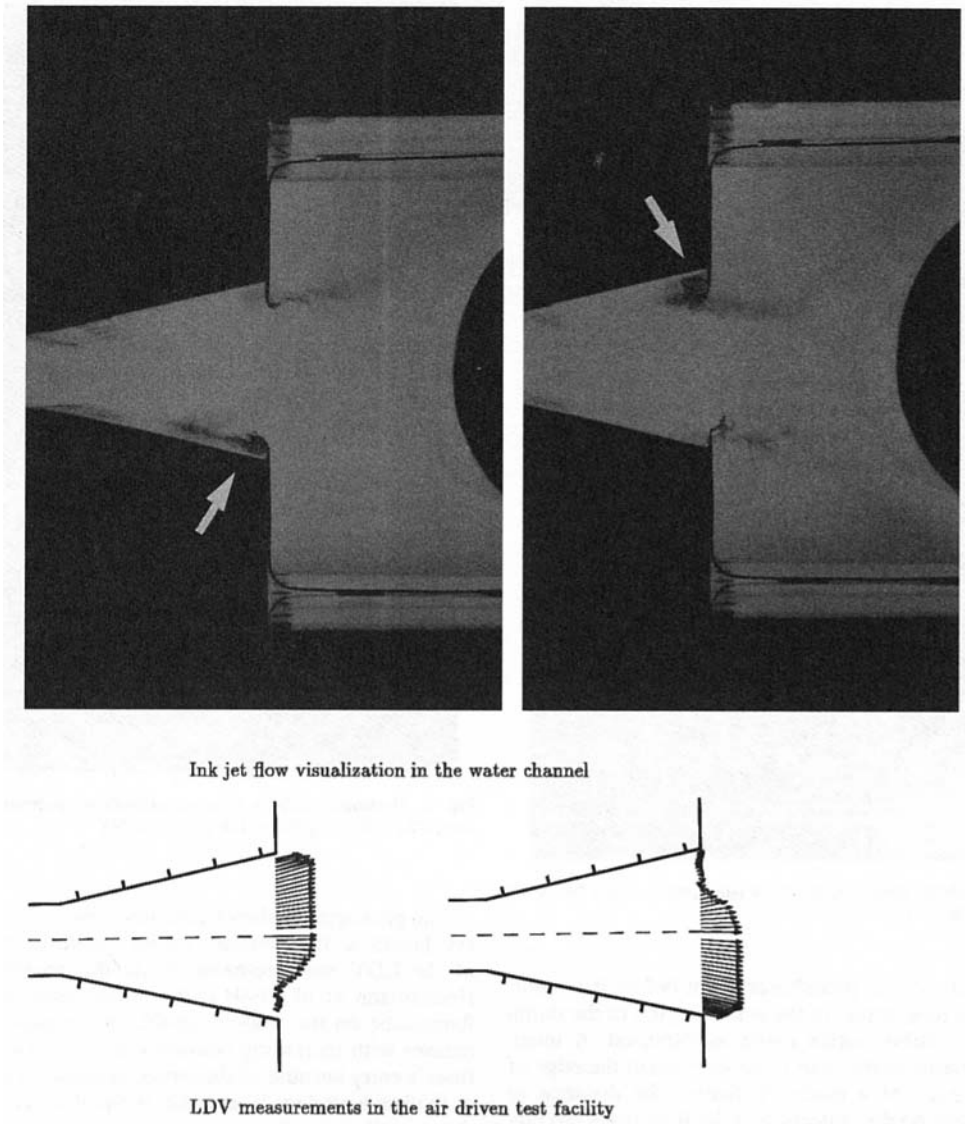

Fig. 10 Occurrence of two similar flow situations in diffusers with large opening angles (geometric data: 2Φ = 25 deg; l/h_1 = 3.0; d/h_1 = 2.51)

possible flow situations may occur with the same probability. Because of the lifetime of these possible flow situations the flow can be regarded as quasi-stationary, but not as a stable flow. The switch from one flow situation to the corresponding mirror-symmetric flow occurs quite stochastically. An impression of this flow is given in Fig. 10 by the visualization of the near-wall flow field applying the ink jet method. In this case, the color was added to the flow at the prediffuser's exit. While in the left-hand photograph the color moves in the upstream direction along the right diverging wall of the prediffuser, viewed in the flow direction, the right-hand photograph shows the opposite effect. Here, backflow occurs along the left diverging wall.

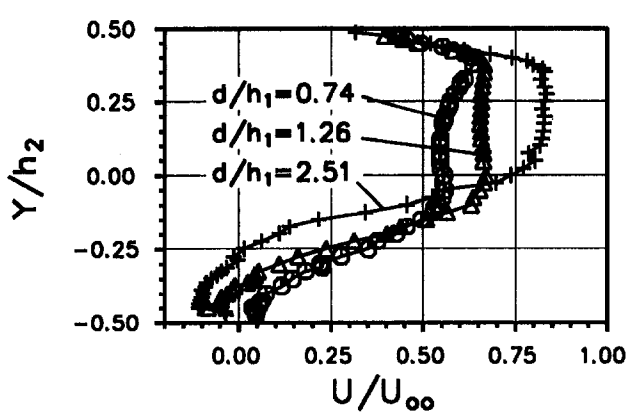

Fig. 11 Measured velocity distribution at the prediffuser's exit at large opening angles (geometric data: 2Φ = 25 deg; l/h_1 = 3.0)

While the flow visualization only is a qualitative description of this phenomenon, it is confirmed by the results of the LDV measurements taken in the air-blown test facility. The vector diagram in Fig. 10 quantitatively shows the mirror symmetry of the two possible flow situations.

At the smallest flame tube distance under examination, d/h_1 = 0.74, the upstream effect of the flame tube causes the flow to reattach near the prediffuser's exit, resulting in a separation bubble inside the prediffuser (see Figs. 9 and 11). Although the flow is attached near the exit of the prediffuser, the velocity profile becomes unsymmetric. Therefore, for prediffuser geometries with a normalized length of l/h_1 = 3.0 and large opening angles, an overall pressure recovery similar to that in combustor diffusers with moderate opening angles can be reached only with very small flame tube distances (see Fig. 12).

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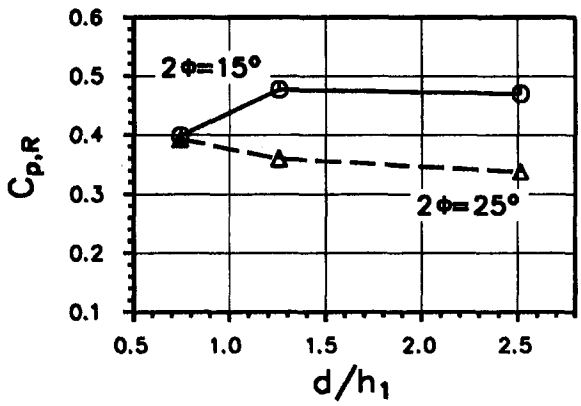

Fig. 12 Experimentally determined overall pressure recovery

Among other parameters influencing the onset of flow separation, the knowledge of the turbulence intensity at the diffuser's entry is of fundamental interest. In the following, some results concerning the influence of turbulence on the pressure recovery attainable in dump diffusers are presented. Elevating the turbulence level increases the momentum exchange between the core flow and the near wall flow. Therefore, the onset of flow separation can be delayed. For example, Stevens and Williams (1980) showed, for an annular diffuser with constant inner diameter, that increasing the turbulence intensity leads to an improvement in flow stability at the diffuser exit. Hoffmann and Gonzalez (1984) showed, for a two-dimensional diffuser with free discharge, that increasing the turbulence level to at least 3.5 percent can improve the pressure recovery drastically. Their data, for example, for a diffuser geometry with a ratio of diffuser length l to inlet height $h_1 = 15$ and an opening angle of $2\Phi = 20$ deg, show an increase in pressure recovery from $C_p = 0.58$ without turbulence generator to $C_p = 0.71$ at increased turbulence level. This improvement in C_p for the condition of increased turbulence intensity is a result of the improved velocity profile with a symmetric velocity distribution because of a delayed flow separation.

The results of our experiments show that this tendency is also valid for combustor dump diffusers. By mounting a turbulence grid consisting of bars with a diameter of $d_b = 8$ mm, the turbulence intensity in the core region was increased from $Tu \approx 1.5$ to $Tu \approx 6$ percent with a nearly identical velocity distribution (see Fig. 13). At small opening angles, where the flow inside the diffuser is attached and remains symmetric even at low turbulence levels, increasing the turbulence intensity results in an increased pressure recovery (see Fig. 14). At larger opening angles, the effect of flame tube distance additionally has to be taken into account; while at small flame tube distances ($d/h_1 = 0.74$) the flow at increased turbulence intensity remains symmetric throughout the prediffuser, the flow becomes unsymmetric at large flame tube distances ($d/h_1 = 2.51$) due to a flow separation despite the increased turbulence intensity (see Fig. 15).

Numerical Calculations

One of the major goals of the research program was to obtain detailed experimental data with well-known boundary conditions to get a better understanding of the complex correlations between diffuser geometry and inlet flow conditions on one hand and flow characteristics in combustor diffusers on the other. Both the air and the water tunnel allow the study of the influencing parameters by isolating the parameters of interest. As a result, a data base is available for the validation of numerical codes. The code used here is a finite volume code solving the time-averaged Navier–Stokes equations on a body-fitted coordinate system first described by Bauer (1989). To take into account the turbulent nature of the flow, the standard $k-\epsilon$ model was used. For all computations the inlet flow conditions as measured two channel heights upstream of the beginning of the

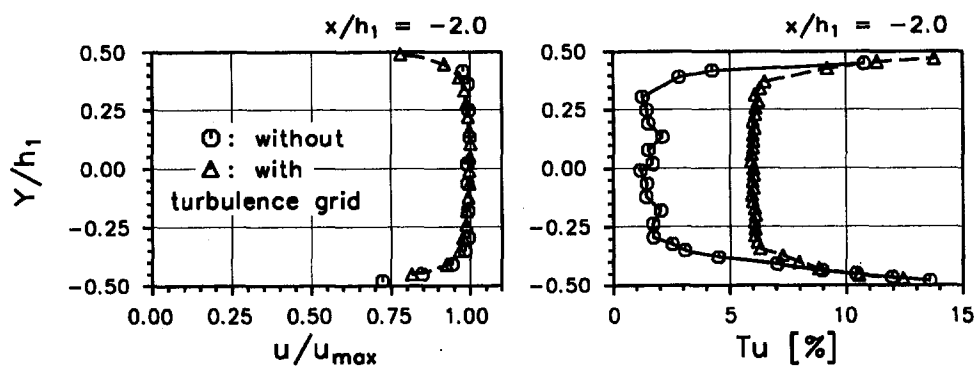

Fig. 13 Velocity and turbulence intensity profiles at the reference position

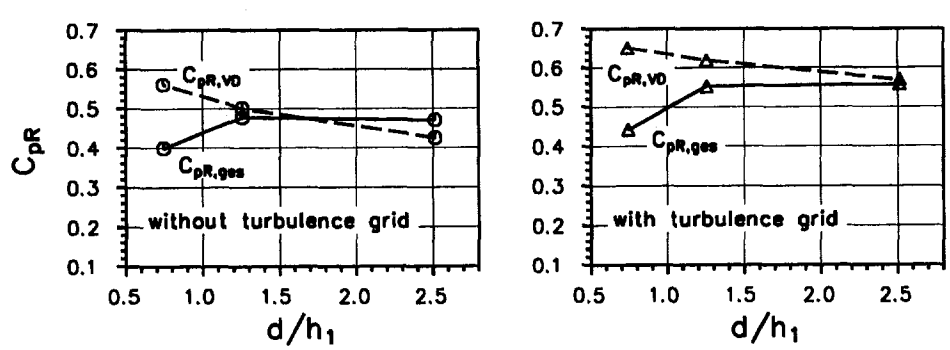

Fig. 14 Influence of turbulence on overall pressure recovery (geometric data: $2\Phi = 15$ deg; $l/h_1 = 3.0$)

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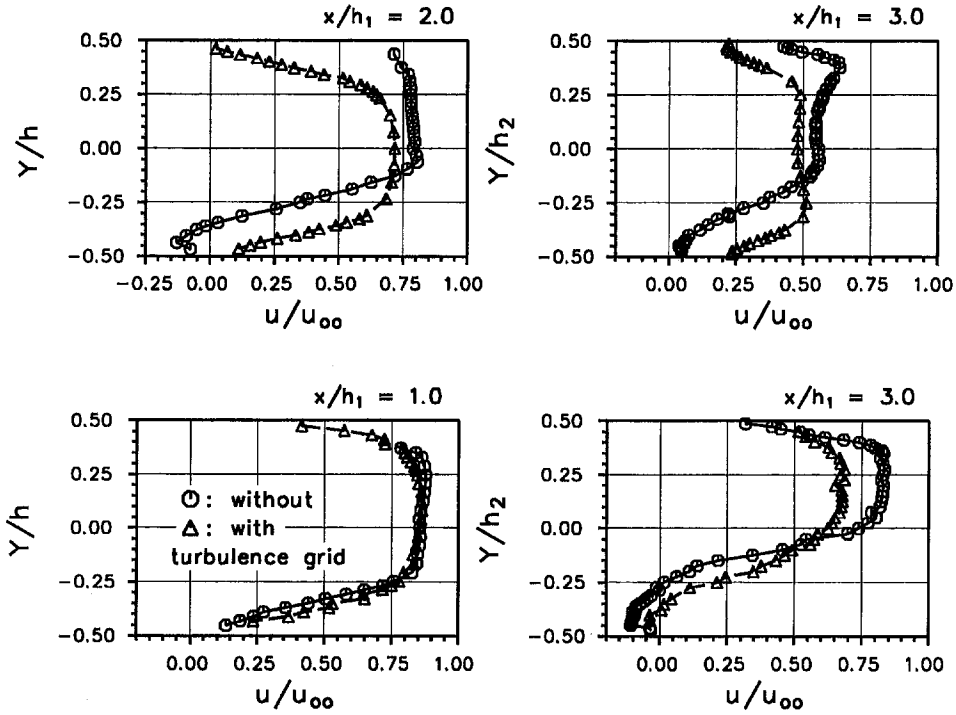

Fig. 15 Influence of turbulence on the velocity distribution inside the prediffuser (geometric data: $2\Phi = 25$ deg; $l/h_1 = 3.0$; upper diagrams: $d/h_1 = 0.74$; lower diagrams: $d/h_1 = 2.51$)

prediffuser have been used as boundary conditions. The length scale was assumed to be 3 percent of the channel height at this reference position. The turbulent kinetic energy was calculated based on the measured fluctuating velocities and based on the assumption of isotropic turbulence. Furthermore, at the exit of the flame tube channels, the gradients were set to zero, and for most of the geometries under examination symmetric conditions were specified along the centerline of the diffuser models for reasons of simplicity.

In the following, the results for a combustor diffuser having a total opening angle of $2\Phi = 15$ deg, a length-to-height ratio of $l/h_1 = 3.0$, resulting in an area ratio of $AR = 1.78$, and a flame tube distance of $d/h_1 = 1.26$ are presented. The grid used for this calculation had 46×30 lines in the x and y directions, respectively. The computed flow field clearly demonstrates excellent agreement between the calculated and the visualized flow field (see Fig. 16). An even better illustration of the capability of the numerical code is the flow field inside the prediffuser. This region is of special importance for the entire flow field in the combustor diffuser. Therefore, a numerical code to predict the flow inside a combustor diffuser should predict the flow in this dominating region with relatively high accuracy. The comparison of the calculated results with the results of the LDV measurements show that the code used here basically is able to fulfill this task. The calculated profiles of the time-averaged velocities shown in Fig. 17 are in excellent agreement with the measured values. The transformation of the velocity profile passing through the prediffuser as well as the upstream effect of the flame tube dome and the relative acceleration of the flow in the near wall region is predicted correctly even though no special attempts were made to discretize the near-wall region. Other effects, for example the correlation between inlet blockage and intensity of the flame tube's upstream effect, were also verified numerically. This is valid for all flame tube distances under examination and for diffuser angles up to $2\Phi = 15$ deg. However, up to now, the phenomenon found in the experiments, that two similar flow situations may occur in diffusers with large opening angles, could not be predicted correctly. Currently, calculations are under way dealing with this problem using an advanced version of the computer code as described in Benz and Wittig (1992). We plan to publish the results of these computations.

Conclusions

Both experimental and numerical results of the flow field in a model combustor diffuser are presented. Emphasis has been

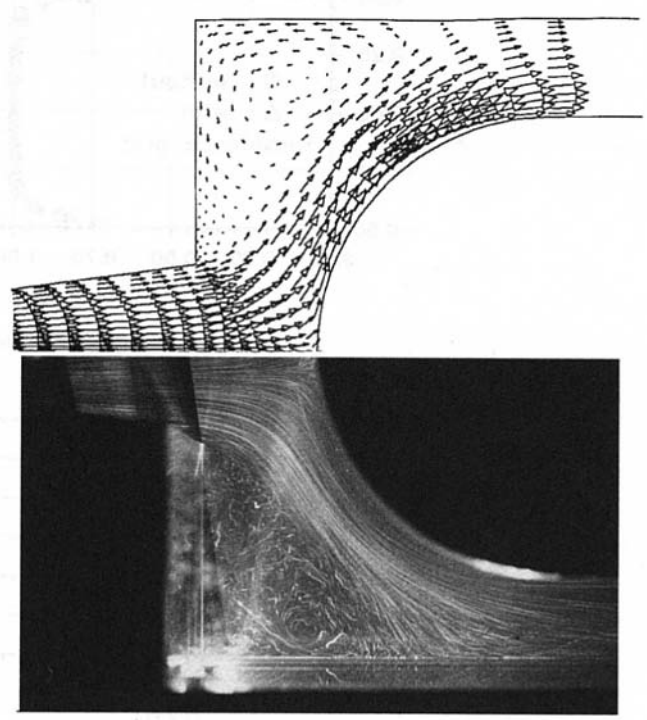

Fig. 16 Comparison of the computed and visualized flow field for a model diffuser with $d/h_1 = 1.26$, $2\Phi = 15$ deg, $l/h_1 = 3.0$

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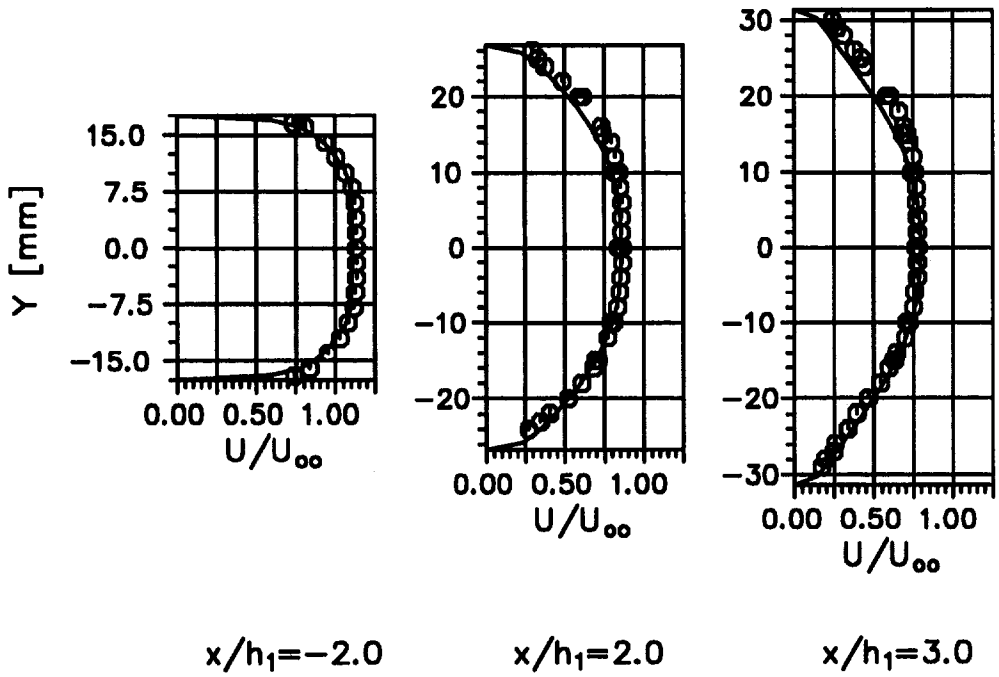

Fig. 17 Measured (= symbols) and calculated (= lines) time averaged axial velocity components inside a prediffuser with d/h_1 = 1.26, 2Φ = 15 deg, l/h_1 = 3.0

directed toward the flow field inside the prediffuser, which is of special importance for the flow field in the entire diffuser–combustor system. Both the experimental and the numerical results show the dominating effect of the flame tube. For a prediffuser length of $l/h_1 \approx 3.0$, flow separation inside the prediffuser can be prevented up to opening angles of about 22 deg by decreasing the flame-tube distance. The largest opening angle under investigation was 25 deg, corresponding to an area ratio of about 2.3. At this opening angle, the flow can be forced to reattach inside the prediffuser only for small flame tube distances, but is strongly distorted at the prediffuser's exit due to a separation bubble inside. At increased turbulence levels at the prediffuser's entry, the flow remains symmetric for small flame tube distances even with large opening angles.

A comparison of the experimental data with the results of a numerical computation demonstrates the capability of predicting the flow field in combustor diffusers with reasonable accuracy, at least for diffusers with an included total opening angle less than 18 deg.

Future work will be directed toward the continuation of our systematic investigations of inlet flow effects and will include the investigation of contoured diffuser walls.

Acknowledgments

The present study was supported by a grant from the Sonderforschungsbereich 167 (high-intensity combustors) of the Deutsche Forschungsgemeinschaft.

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